

Weekly Progress Report

Refactoring Detection Project - Week 5 progress update for the Monday research supervisor meeting

1. Overview

During Week 5, the team moved from UI design and planning into active implementation. Each member worked on a specific part of the AntiCopyPaster UI Panel, using separate branches and preparing changes for review through pull requests into the ui_panel branch. The focus was no longer only on prototypes; the team began translating Figma and design decisions into working Java Swing UI code inside the IntelliJ plugin.

This week included implementation progress on the multiple clone popup, diff preview, AI Trust / validation interface, buttons and actions, and the explanation section. The team also continued testing, reviewing branches, helping resolve conflicts, and preparing weekly website files.

2. Weekly Progress Summary

Area	Progress Made
Implementation	All team members worked on updating their assigned UI sections in the code instead of only planning the design.
Branches	Team members continued working on separate branches. When a section is ready, the branch is pushed and a pull request is opened into ui_panel.
Review	The team planned to assign reviewers for pull requests so code can be checked before merging.
Merge process	Accepted pull requests will be merged into the shared ui_panel branch after review and conflict resolution.
Website/files	Weekly files continued to be organized for the website, including slides, report, collaborators, results, and related progress documents.

3. Meeting Context and Week 5 Focus

The Week 5 slides show that the team is now in the implementation stage. The main project workflow is to update each assigned UI section on a separate branch, open pull requests into ui_panel when ready, review the changes, and then merge accepted work into the shared UI branch.

The implementation work follows the task division from Week 4: Jake worked on the multiple clone UI, Amaro worked on the explanation section, Juliana worked on the diff preview, Dhir worked on AI Trust / validation, and Fintan worked on buttons and actions. The Week 5 work also included collaboration across branches, testing, conflict help, and review preparation.

4. Individual Implementation Updates

Team Member	Week 5 Contribution
Juliana	Added final details to the Figma Diff Viewer and began developing the UI Panel on her Diff-Preview_UI branch. She researched Java Swing components, learned from the existing UI Panel, and started building a cleaner Diff Viewer Panel. She replaced the code editor/viewer approach with card-based code sections so the original duplicate code and proposed refactoring are easier to read.
Dhir	Turned the AI Trust prototype into a working feature in RefactoringSuggestionPanel.java. He implemented a compact and collapsible AI Trust interface with the overall score, trust level, five equally weighted validation components, and progress indicators. He tested success, verifying, and failure states in the IntelliJ sandbox, restored the production version after removing mock controls, verified Java 17 and Gradle compilation, committed the feature to his AI_trust branch, pushed through a fork, opened Pull Request #30 into ui_panel, and recorded a demo for the team presentation.
Jake	Coded and tested the multiple clone popup. His work focused on the case where more than one clone is detected, so the user can review clone occurrences separately and decide whether to include or exclude each one before refactoring.
Fintan	Implemented changes to the buttons UI, including changes connected to user actions and validation behavior. He also helped teammates with individual implementation issues that came up during the week.

Team Member	Week 5 Contribution
Amaro	Continued the weekly website and documentation work, including organizing the files for the project website. He also worked on the explanation section, merged the explanation-section work, tested Dhir's AI Trust code, and helped other team members with merging and resolving conflicts. His work supported both the UI implementation and the weekly progress organization.

5. UI Sections Implemented or Advanced

UI Section	Week 5 Progress	Why It Matters
Multiple Clone UI	Jake coded and tested the popup used when the tool detects multiple clones. The design keeps the separate-card idea and gives the user a choice for each clone occurrence.	Developers should not be forced to refactor every clone at once. Each clone should be reviewed and accepted or excluded individually.
Explanation Section	Amaro improved and merged the explanation section. The section focuses on clone type, file location, pasted code location, line numbers, method names, selected ranges, and suggested method.	The developer needs to quickly understand what was detected, where it is located, and what Extract Method refactoring is being suggested.
Diff Preview	Juliana started developing the Diff Viewer Panel using Java Swing components and a card-based layout based on the Figma design.	A cleaner diff preview makes the original duplicate code and proposed replacement easier to compare before the user applies changes.
AI Trust / Validation	Dhir implemented the AI Trust interface as a compact and collapsible section with score, trust level, validation components, and multiple states.	The AI Trust section explains whether clone detection, refactoring, usefulness, compilation, and testing evidence support the suggestion.
Buttons / Actions	Fintan implemented button UI changes, including behavior around direct editing and disabled acceptance when validation fails.	Actions need to match the validation state so users do not apply incomplete or failed refactorings.

6. Multiple Clone UI Progress

The multiple clone UI continued from the Week 4 decision that each clone should appear separately. Week 5 work moved that decision into code by building and testing the multiple clone popup. The popup supports the idea that a user should be able to inspect each clone occurrence and choose whether to include it in the refactoring.

- Separate clone cards or occurrences keep the interface organized when more than one clone is detected.
- The user can make an individual choice for each clone instead of accepting all results at once.
- Include and Exclude actions support clearer control over what should be refactored.

7. Diff Preview Progress

The diff preview work moved from Figma into the UI Panel code. Juliana researched Java Swing because the existing panel uses Swing components, then began implementing the Diff Viewer Panel on a separate branch. The goal is to replace a harder-to-read code editor/viewer with cleaner code cards for the original duplicate code and extracted replacement.

- The new layout is based on the original panel, the Figma Diff Viewer design, and the refactoring suggestion prototype.
- The card-based design is intended to make the code comparison easier to scan.
- This section supports the developer decision process before applying the refactoring.

8. AI Trust / Validation Progress

The AI Trust section made major implementation progress. Dhir converted the prototype into a working feature inside RefactoringSuggestionPanel.java. The section is compact and collapsible so it can show important validation information without taking too much space from the diff viewer.

Feature	Implementation Meaning
Overall AI Trust score	Shows the high-level trust result for the refactoring suggestion.
Trust level	Gives a simple label that summarizes the current validation confidence.

Feature	Implementation Meaning
Five validation components	Shows Clone Detection, Refactoring Agent, Usefulness, Compilation, and Test / Behavior as equally weighted components.
Progress indicators	Helps show whether validation is verified, still running, or failed.
Validation states	Supports successful, verifying, and failure states so the UI can explain what is happening.

9. Buttons and User Actions Progress

The buttons UI work focused on making the action area more useful and safer. The slides show two main ideas: direct editing and disabled acceptance. Direct editing gives the user a way to edit the proposed code before applying it. Disabled acceptance keeps Apply unavailable when validation fails or when there is nothing safe to apply.

- Apply should not be enabled when validation fails.
- Regenerate, Cancel, and edit-related actions should support the user without forcing a bad change.
- Button behavior should clearly reflect the current workflow state.

10. Explanation Section Progress

Although the explanation section was not shown as a separate Week 5 slide, it was part of the implementation work. Amaro worked on making the explanation clearer and more useful for developers. This section is important because it is the first place users look to understand what the tool found and what it wants to change.

Information to Show	Purpose
Clone type	Explains whether the clone is Type-1 or another clone category.
File locations	Separates existing clone location from pasted code location.
Line numbers and methods	Makes it easier to jump to the exact code that was detected.
Suggested method	Explains the Extract Method suggestion in simple terms.
Detection reason	Tells the developer why the code was considered duplicated.

11. Branch, Review, and Merge Workflow

The Week 5 workflow is centered on branches and pull requests. Team members work on their own branches, push their work, and open pull requests into the ui_panel branch. The team will then review each pull request, fix conflicts if needed, and merge accepted work into ui_panel.

- Dhir opened Pull Request #30 for the AI Trust scoring and validation UI.
- Other members continued preparing or testing their branches for later review.
- Amaro helped with merging and conflict resolution so individual UI sections can be combined more smoothly.

12. Next Steps

- Finish each assigned UI section and push the branch to GitHub.
- Open pull requests into the ui_panel branch when each section is ready.
- Assign reviewers for each pull request and check the code before merging.
- Fix conflicts between sections, especially where multiple changes touch RefactoringSuggestionPanel.java.
- Continue testing the plugin in the IntelliJ sandbox and sharing screenshots or demos with the team.
- Keep updating the website files so the weekly progress remains organized.

13. Conclusion

Overall, Week 5 focused on implementation. The team moved from finished Figma designs and UI plans into working code, branch management, testing, pull requests, and merge support. Each member made progress on a specific UI section, and the project is now closer to having a complete IntelliJ-style AntiCopyPaster UI Panel with clearer explanations, better diff previews,

AI Trust visibility, safer actions, and support for multiple clone cases.

14. Slide Evidence Used for This Report

Slide Area	Report Connection
Weekly Progress Summary	Used to summarize implementation, branch, review, and merge progress.
Multiple Clone UI	Used to describe separate cards, individual user choice, and include/exclude actions.
Diff Preview	Used to describe the card-based diff viewer direction and cleaner comparison layout.
AI Trust Demo / PR #30	Used to describe the completed AI Trust implementation and pull request for team review.
Buttons	Used to describe direct editing and disabled Apply behavior when validation fails.
Next Steps	Used to summarize pull requests, team review, conflict fixing, merging, and website updates.